Preg Assist

25-26J-253

01

Problem Statement & Stakeholders

- **Context:** Prenatal healthcare across hospitals, maternity clinics, and resource-limited rural communities.
- **Stakeholders:** Pregnant mothers [Primary Users], OBGYNs & remote healthcare workers [Secondary Users], and Hospital administrations [Decision Makers].
- **Observed Issue:** Prenatal care is restricted to episodic clinic visits and fragmented tools, causing delayed detection of critical physical, mental, and fetal risks.
- **Why it Matters:** Delayed intervention increases severe maternal and fetal risks. An integrated system reduces these risks, prevents clinical errors, and bridges accessibility gaps.

Research Gap & Objectives

02

- **Gap:** Current solutions focus on isolated medical data and episodic clinic visits, failing to provide continuous, integrated monitoring of maternal and fetal health.
- **Main objective:** To develop an integrated prenatal care platform that unifies physical/mental health tracking, AR education, and explainable AI to reduce maternal and fetal risks through continuous support.

03 Solution Overview

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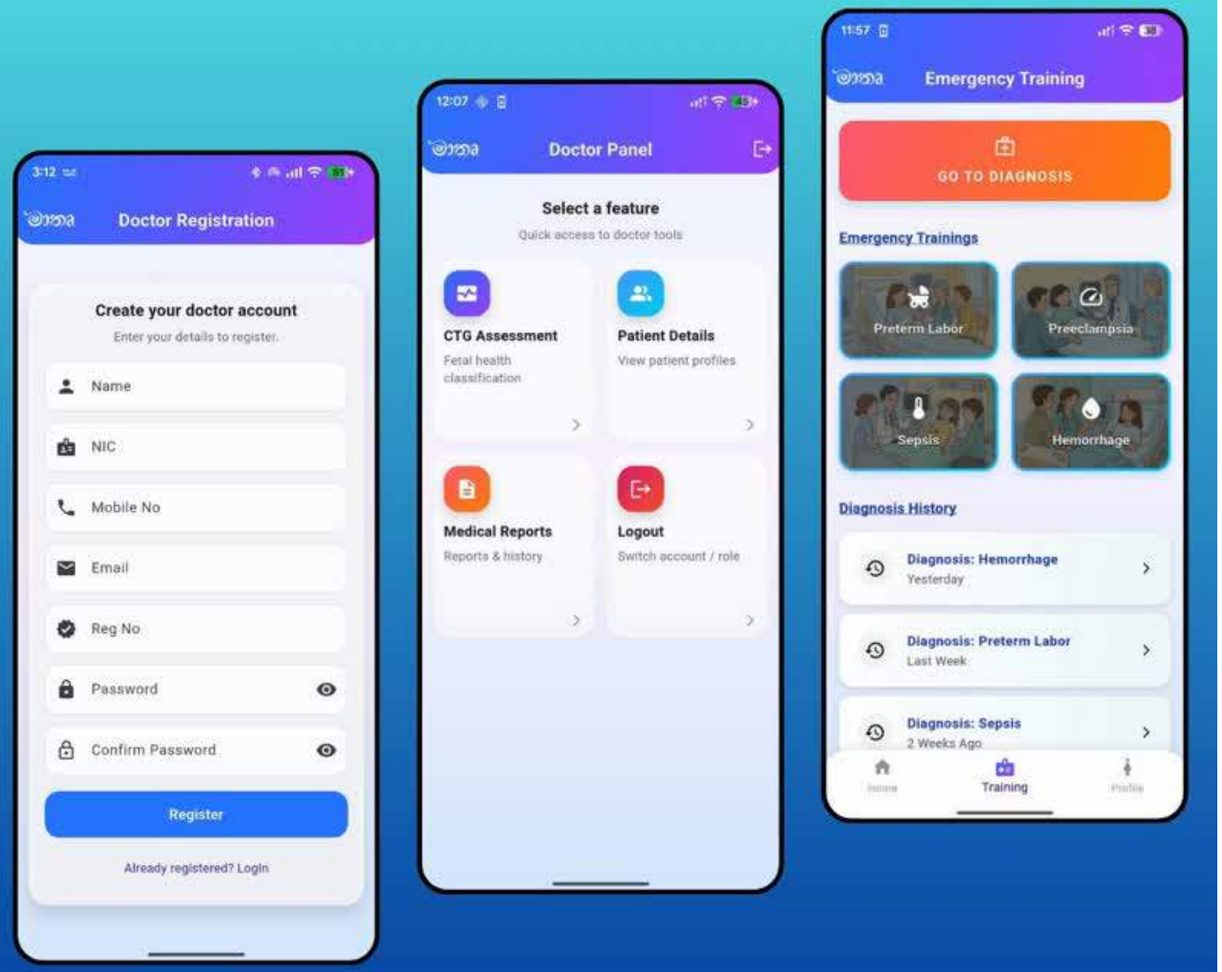
Intelligent support for mothers & clinicians



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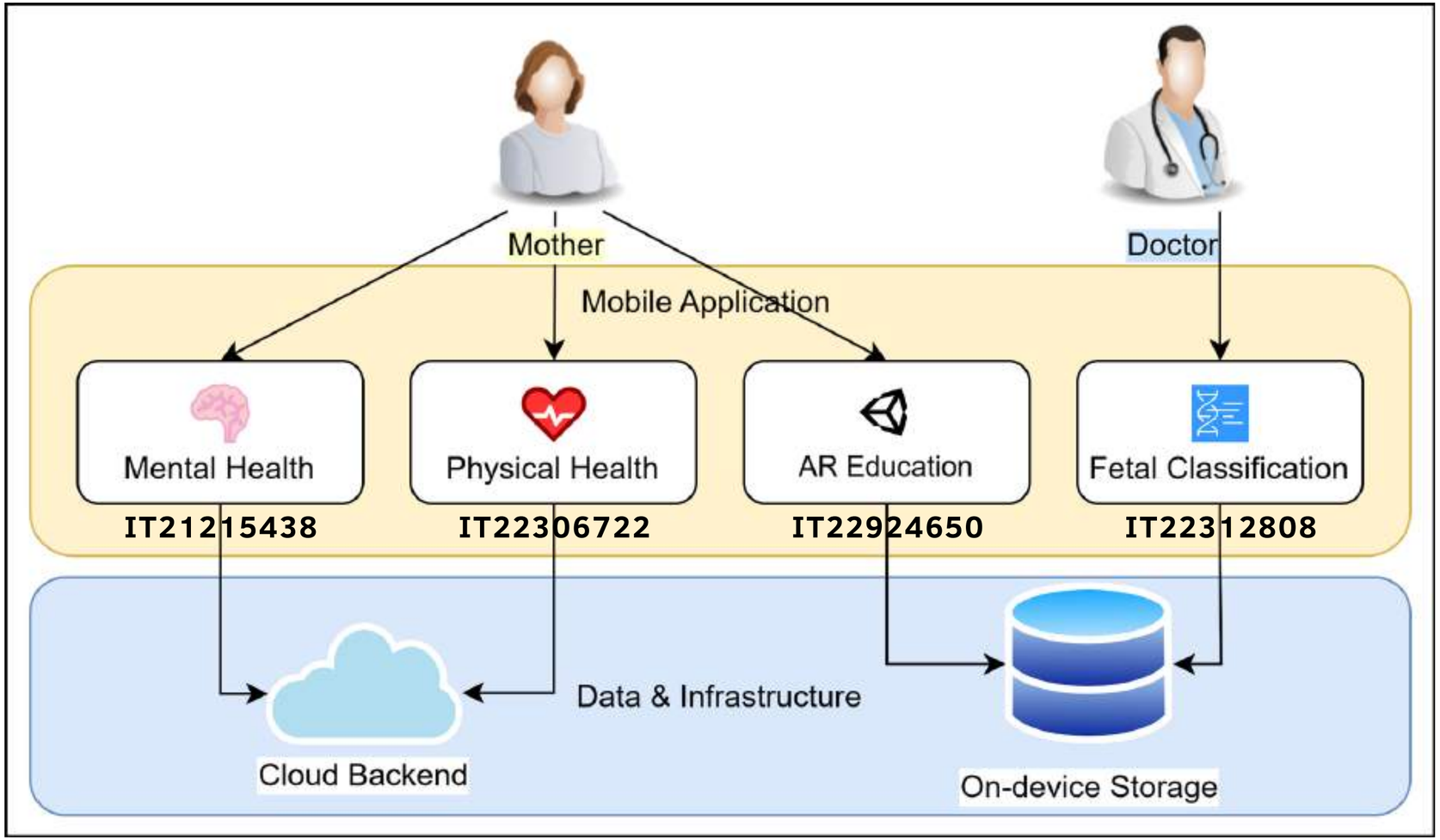
System Architecture & Integration



Integration Points:

- Flutter-to-Unity bridge (CLEAR_TASK protocol) for AR.
- On-device Edge AI inference.
- REST APIs / JSON for secure Cloud Backend sync.

Deployment Setup:

- Mobile-first edge deployment (Smartphones) paired with a centralized Cloud Backend for federated learning.

05 Commercialization & Impact

- **Target users/market:** Pregnant mothers, Gynecologist, and healthcare institutions in both urban and resource-limited settings.
- **Value proposition:** An all-in-one, mobile-first platform combining explainable AI, AR education, and privacy-aware on-device processing.
- **Cost and feasibility:** Freemium access for mothers, premium clinical features for professionals, and B2B hospital partnerships.
- **Adoption plan / next steps:** Transition from successful pilot demonstrations to direct hospital system integrations, leveraging high smartphone penetration.

Demo Plan

06

Demo flow (end-to-end):

Decision Support System for Fetal Health Classification → AR and AI-Driven Module for Personalized Pregnancy Emergency Training → AI-Based Mental Health Monitoring and Risk Prediction → Physical Health Monitoring and Wellness guidance.

Where each member takes over:

- **B.P Buddika (IT22312808)** → Decision Support System for Fetal Health Classification
- **Jayasinghe N. (IT22924650)** → AR and AI-Driven Module for Personalized Pregnancy Emergency Training
- **Malikshi Perera (IT21215438)** → AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women
- **Wijethunga D.N (IT22306722)** → Physical Health Monitoring and Wellness guidance for Pregnant Mothers

Thank you!

Feel free to ask us questions if you have any.